Practical No. 2

#c language

```
#include <stdio.h>

int main()

{

   int a[100], n, i, j, key;

   int linear = -1, binary = -1;

   int low, high, mid, temp;


   printf("Enter number of customer IDs: ");

   scanf("%d", &n);


   printf("Enter customer account IDs:\n");

   for (i = 0; i < n; i++)

   {

      scanf("%d", &a[i]);

   }


   printf("Enter customer ID to search: ");

   scanf("%d", &key);


   // Linear Search

   for (i = 0; i < n; i++)

   {

      if (a[i] == key)

      {

         linear = i;
```

```
      break;
    }
}

if (linear != -1)
    printf("\nLinear Search: ID found at position %d\n",
          linear + 1);
else
    printf("\nLinear Search: ID not found\n");

// Sort the array for Binary Search
for (i = 0; i < n - 1; i++)
{
    for (j = 0; j < n - i - 1; j++)
    {
        if (a[j] > a[j + 1])
        {
            temp = a[j];
            a[j] = a[j + 1];
            a[j + 1] = temp;
        }
    }
}

// Binary Search
low = 0;
```

```
high = n - 1;

while (low <= high)
{
   mid = low + (high - low) / 2;

   if (a[mid] == key)
   {
      binary = mid;
      break;
   }
   else if (a[mid] < key)
   {
      low = mid + 1;
   }
   else
   {
      high = mid - 1;
   }
}

if (binary != -1)
   printf("Binary Search: ID found at position %d in sorted list\n",
        binary + 1);
else
   printf("Binary Search: ID not found\n");
```

```
    printf("\nTime Complexity:\n");
    printf("Linear Search: O(n)\n");
    printf("Binary Search: O(log n)\n");

    return 0;
}
```

Sample Output

Enter number of customer IDs: 5

Enter customer account IDs:

105 102 108 101 106

Enter customer ID to search: 108

Linear Search: ID found at position 3

Binary Search: ID found at position 5 in sorted list

Time Complexity:

Linear Search: O(n)

Binary Search: O(log n)